Introduction to LaTeX

Saarvik Singh Suryavanshi

2025UCP1681

Course: Technical Writing

Course Coordinator: Neeta Nain

Abstract

This report introduces LaTeX and explains how it can be used to prepare well-formatted documents. It covers the basic document structure, commonly used commands, mathematical expressions, and the use of tables and figures. The aim is to understand the basic features of LaTeX and how they help in preparing academic documents.

0.1 Introduction

LaTeX is a document preparation system commonly used for writing academic papers, reports, and technical documents. It is especially useful when a document contains mathematical expressions, tables, references, or figures.

Unlike a normal word processor, LaTeX uses commands to describe the structure and formatting of a document. This makes it easier to maintain consistent formatting throughout a document.

Chapter 1

Introduction to LaTeX

1.1 What is LaTeX?

LaTeX is built on the TeX typesetting system. It allows users to focus on the content while the system handles most of the formatting.

Some common uses of LaTeX are:

- Writing technical and research papers.
- Preparing project reports and assignments.
- Creating documents with mathematical equations.
- Preparing presentations.

1.2 Advantages of LaTeX

LaTeX provides several useful features:

- Consistent formatting throughout the document.
- Good support for mathematical equations.
- Automatic numbering of sections, tables, and figures.
- Easy management of references and citations.

Chapter 2

Basic Document Structure

2.1 Document Class

Every LaTeX document begins with the `\documentclass` command.

It specifies the type of document being prepared. Some commonly used document classes are:

- article
- report
- book
- beamer

2.2 Basic Commands

A LaTeX document generally contains a preamble and a document environment.

The preamble contains the document class and packages. The actual content is written between `\begin{document}` and `\end{document}`.

2.2.1 Sections and Subsections

Sections are used to divide a document into smaller parts. Subsections help organize the content further.

Chapter 3

Mathematics, Tables and Figures

3.1 Mathematical Expressions

LaTeX is useful for writing mathematical expressions. For example, the quadratic formula can be written as:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Equations can also be numbered and referenced in a document.

3.2 Tables and Figures

Tables are used to present information in rows and columns. Figures can be used to include images, graphs, and diagrams.

Both can be given captions and labels, which make them easier to identify and reference in a report.

Chapter 4

Conclusion

LaTeX is a useful tool for preparing academic and technical documents. Its structured approach makes formatting easier to manage, especially in longer documents.

Learning its basic commands, environments, and packages provides a good starting point for creating well-organized reports and presentations.